RAJALAKSHMI ENGINEERING COLLEGE

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S2

Started on	Monday, 31 August 2026, 9:58 PM
State	Finished
Completed on	Monday, 31 August 2026, 9:58 PM
Time taken	48 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int floorValue(int a[],int low,int high,int x){
    if(low>high) return -1;
    int mid=(low+high)/2;
    if(a[mid]==x) return a[mid];
    if(a[mid]<x){
        int res=floorValue(a,mid+1,high,x);
        return (res==-1)?a[mid]:res;
    }
    return floorValue(a,low,mid-1,x);
}

int main(){
    int n,x;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++) scanf("%d",&a[i]);
    scanf("%d",&x);
    printf("%d",floorValue(a,0,n-1,x));
    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔

	Input	Expected	Got	
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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